

# Questions on the manuscript and the talk

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Four questions, each answered from an artifact in the repository rather than from recollection. Sections 1 and 2 concern one sentence of section 4 of the manuscript; sections 3 and 4 concern one sentence of the speaker notes. What was *not* checked is stated at the end of each answer and collected under Provenance.

## 1 What does 356 mean, and are the couplings equispaced?

The sentence at issue opens section 4, *Results*:

“The dataset contains 356 accepted solutions at 13 couplings from  $\alpha = 0$  to  $\alpha = 1/2$  in units  $r_H = 1$ .”

### The 356 is a total over couplings, not a count per coupling

It is the number of converged black-hole solutions *summed over all thirteen couplings* — not 356 per coupling, and nothing to do with the number of  $\alpha$  values. Each coupling carries one *walk*: a continuation in the spin at that fixed  $\alpha$  (section 2 below). The number 356 is the total length of the thirteen walks.

| $\alpha$     | states     | walk   | $TJ^{1/3}$ range                               |
|--------------|------------|--------|------------------------------------------------|
| 0            | 43         | full   | $1.12 \times 10^{-5}$ to $8.74 \times 10^{-2}$ |
| 0.005        | 10         | narrow | $2.44 \times 10^{-3}$ to $5.10 \times 10^{-2}$ |
| 0.01         | 11         | narrow | $1.21 \times 10^{-3}$ to $5.14 \times 10^{-2}$ |
| 0.02         | 40         | full   | $4.61 \times 10^{-3}$ to $8.15 \times 10^{-2}$ |
| 0.035        | 11         | narrow | $3.76 \times 10^{-3}$ to $5.13 \times 10^{-2}$ |
| 0.05         | 45         | full   | $4.49 \times 10^{-4}$ to $7.42 \times 10^{-2}$ |
| 0.075        | 16         | narrow | $4.38 \times 10^{-4}$ to $4.75 \times 10^{-2}$ |
| 0.1          | 45         | full   | $3.84 \times 10^{-4}$ to $6.48 \times 10^{-2}$ |
| 0.15         | 17         | narrow | $2.57 \times 10^{-4}$ to $3.90 \times 10^{-2}$ |
| 0.2          | 45         | full   | $1.44 \times 10^{-4}$ to $5.21 \times 10^{-2}$ |
| 0.3          | 16         | narrow | $1.06 \times 10^{-4}$ to $2.59 \times 10^{-2}$ |
| 0.4          | 13         | narrow | $5.06 \times 10^{-5}$ to $1.97 \times 10^{-2}$ |
| 0.5          | 44         | full   | $3.34 \times 10^{-6}$ to $3.34 \times 10^{-2}$ |
| <b>total</b> | <b>356</b> |        |                                                |

Table 1: States per coupling, read from `results/egb-extremality.json`. The two populations are the two kinds of walk described in section 2.

The counts are bimodal, and the split is exactly the one the next sentence of section 4 describes. The six couplings walked over the *whole* spin range,  $\alpha = 0, 0.02, 0.05, 0.1, 0.2, 0.5$ ,

carry 40–45 states each, 262 in total. The seven that start at  $q = 0.6$  and run only to the near-extremal end carry 10–17 each, 94 in total. So

$$356 = \underbrace{262}_{6 \text{ full walks}} + \underbrace{94}_{7 \text{ narrow walks}}.$$

“Accepted” means the state passed the solver’s acceptance gate: the stored rows carry the diagnostics it is judged on (`max_relative_tensor_residual`, `psi_window_spread`, `mass_tail_spread`, and so on). Refused attempts are not in the count.

## The couplings are not equispaced

|          |   |      |      |     |      |      |      |      |     |     |    |    |    |
|----------|---|------|------|-----|------|------|------|------|-----|-----|----|----|----|
| $\alpha$ | 0 | .005 | .01  | .02 | .035 | .05  | .075 | .1   | .15 | .2  | .3 | .4 | .5 |
| gap      |   | .005 | .005 | .01 | .015 | .015 | .025 | .025 | .05 | .05 | .1 | .1 | .1 |

The spacing is graded, widening by a factor of about 20 from the small- $\alpha$  end to the large- $\alpha$  end. That matches what is being measured: the result is the departure from the first-order value

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2),$$

so the  $\alpha \rightarrow 0$  end has to be resolved finely enough to see the slope leave  $\pi$ , while the large- $\alpha$  end only has to track a slowly varying curve.

The grid is also aligned with its two uses. The six full walks sit at 0, 0.02, 0.05, 0.1, 0.2, 0.5, and the seven narrow ones interleave between them. The narrow walks exist only to add resolution to the extremal-slope curve, which is why they skip the low-spin region entirely.

## 2 What is a “walk”?

A *walk* is one continuation sequence: a chain of solutions at *fixed*  $\alpha$ , stepped in the spin, each solve seeded by the previous one. It is the unit the dataset is organised into — thirteen couplings, thirteen walks, 356 states. What follows is read from `src/rotating_bh/extremality.py`.

### Where a walk starts

`walk_to_extremality(alpha_gb)` first builds a base solution with `predictor_ladder(start=.60, ...)`: it climbs in the *coupling* from the known  $\alpha = 0$  Myers–Perry solution up to the target  $\alpha$  in steps of `coupling_step=.005`, holding the spin at  $\Omega_H = 0.6$ . In units  $r_H = 1$  one has  $q = r_H\Omega_H = \Omega_H$ , which is why the manuscript calls this  $q = 0.6$ .

### Where a walk goes

From that base it runs `walk` twice, in both directions in the spin, with `spin_step=.01` in  $\Omega_H$ :

- `direction=-1`, down to  $\Omega_H = 0$ , the static limit;
- `direction=+1`, up until the temperature falls below `floor=1e-6`, the near-extremal end.

The walk is the concatenation `descending + [anchor] + ascending`, sorted in  $\Omega_H$ . Its length is that coupling’s row in the table above. This is precisely the six-versus-seven split: the couplings given both directions are the 40–45-state full walks, while the seven narrow ones were given only the ascending branch from  $q = 0.6$  — the region the  $T \rightarrow 0$  extrapolation actually consumes.

## Why it is a walk and not a scan

Every step is Newton-solved from the previous solution as its seed. These states cannot be solved independently, because near extremality there is no good initial guess except a neighbour. Three behaviours of the loop follow from that, each recorded in the source as the answer to a way an earlier version failed:

- **Halve, do not stop.** A Newton failure is treated as a seeding accident rather than a physical boundary: the step halves and retries. What ends a walk is the step collapsing below `minimum_step=1e-5`. The docstring puts it plainly — “a wall which looked physical was not”.
- **Roll back on a refused state.** A state that misses the acceptance gate is not kept as the seed. Otherwise every later solve inherits the bad profile and the walk dies far short of where it could reach, which the source records as what the first production run did at the two smallest couplings.
- **Raise the resolution before shortening the step.** Approaching extremality the independent tensor residual grows, and the cure is  $N = 64 \rightarrow 96$ , not a smaller step. The resolution that accepted each state is stored with it, which is the `resolution` field in the rows.

## Why 356 is not simply steps times couplings

A walk steps far more often than it records. States are kept on a schedule: whenever the spin has moved by `spin_record=.02` or the temperature has changed by a factor of `temperature_record=1.3`. Measuring at every accepted step would spend the run in `diagnose`; measuring on a fixed spin grid would crowd the samples where nothing changes. So 356 counts *recorded*, *accepted* states, and the number of Newton solves behind them is substantially larger.

This also explains the temperature coverage in the table of section 1. The  $1.3\times$  temperature trigger makes the samples bunch up automatically as  $T$  collapses towards extremality — which is where `extremal_state` needs them, since it fits only the `window=12` coldest states.

## 3 What is the “rotating solver”?

The sentence at issue is in the speaker notes for the cost slide:

“Reproducing all known solutions took about four hours; the first new object, the rotating solver, took five times more ...”

It is the code that finds rotating Einstein–Gauss–Bonnet black holes: `src/rotating_bh/egb_rotating_bvp.py`, built in milestone (*hito*) 4. The ten-minute notes say it at greater length — “the first genuinely new object, the rotating Gauss–Bonnet black hole” — which is what the five-minute line compresses.

### What it is

A spectral collocation solver for the radial boundary-value problem of the rotating EGB system. From its docstring:

- Chebyshev–Lobatto nodes; damped Newton with a *complex-step* Jacobian and a line search — the same architecture as the vacuum solver `vacuum_bvp.py`, deliberately mirrored.

- Four compact amplitudes  $B, F, H, W$  on a compactified radius,  $z = 1/r$  and  $x = 1 - z$ , so that  $x = 0$  is the horizon  $r_H = 1$  and  $x = 1$  is  $r \rightarrow \infty$ .
- The horizon row is singular in the generic interior formula — the coefficient matrix of the underlying  $[f', b'', h'', w'']$  system degenerates there — so it uses separately derived horizon relations, verified against the exact Myers–Perry closed form.
- At infinity,  $B = 1, F = 1, H_x = 0, W_x = 0$ .
- At  $\alpha_{\text{GB}} = 0$  the interior equations reproduce the vacuum ones exactly.

Everything in the paper rests on it: the walk of section 2 calls into it at every step, and the 356 states are its accepted outputs.

### Why it was the expensive block

Hito 4 cost 11.2 hours and 508 M tokens, 31% of the project’s time and 36% of its tokens — the largest single block. The “five times” of the notes is a token ratio, and it is a comparison across different units:

| Block                                       | Hours       | Tokens             |
|---------------------------------------------|-------------|--------------------|
| Hitos 0–3, reproducing the known literature | 4.4         | 101,558,821        |
| <b>Hito 4, rotating EGB black hole</b>      | <b>11.2</b> | <b>507,954,588</b> |

$508/101.6 = 5.0$ . Note that the sentence quotes hours for the first block and tokens for the second.

The ledger documents why it was costly:

- **The direct tensor derivation was symbolically intractable.** Christoffel and Riemann for the non-diagonal metric did not terminate. The work moved to the reduced action, with equivalence to the full tensor verified *numerically* over generic profiles rather than symbolically.
- **Five discarded solver routes** before the one that worked: fitting a single horizon parameter; multi-parameter shooting with `least_squares` and `solve_ivp`; `solve_bvp` seeded from Myers–Perry with an exact analytic Jacobian, which went singular with parameters diverging to  $10^4$ – $10^{22}$ ; extracting the horizon conditions from the compactified form, where `sp.limit` did not terminate; and translating the raw relations into compact variables, which gave correct but unrecognisable expressions that failed to reproduce even the vacuum form. The route that worked redid the horizon and infinity expansions *directly* in the compact variables.
- **Three self-inflicted errors**, each costed:  $z$  and  $x$  treated as independent symbols; the numerator extracted before the series, producing a spurious relation; and  $f'', w''$  omitted from the physical jets, which left an order-one tensor residual even on exact Myers–Perry.

## 4 Is it a different solver from the one of arXiv:1010.0860, and what was it used to reproduce there?

Different in method and independently written. The second half of the question has the more interesting answer: the rotating solver was never successfully checked against their *bulk numbers at finite coupling*. The single comparison that would do that is the one that failed.

## The two solvers

|                 | Brihaye, Kleihaus, Kunz, Radu<br>arXiv:1010.0860                                               | This project<br>egb_rotating_bvp.py                                             |
|-----------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Method          | COLSYS, a standard off-the-shelf BVP package (their ref. [41]: Ascher, Christiansen & Russell) | Chebyshev–Lobatto spectral collocation, written from scratch                    |
| Nonlinear solve | Newton–Raphson with adaptive mesh selection                                                    | Damped Newton, complex-step Jacobian, line search                               |
| Grid            | $10^2$ – $10^3$ mesh points, adaptive                                                          | $N = 64$ , raised to 96 on a refused state                                      |
| Quoted accuracy | relative $\sim 10^{-6}$                                                                        | tolerance $10^{-11}$ , with per-state residual gates                            |
| Equations       | their own reduction                                                                            | independently rederived via the reduced action, equivalence checked numerically |

What the two share is a strategy, not code. They also continued from the closed-form solutions, “increasing gradually  $\alpha$  or  $\Omega_H$ ”, which is structurally the walk of section 2 together with the climb in the coupling that seeds it.

## What was actually reproduced against them

The eight external contrasts of `results/egb-rotating-external-contrasts.json`, with their sectors, which are the point:

| Check                                               | Sector         | Result                |
|-----------------------------------------------------|----------------|-----------------------|
| reduced normalisation pinned                        | convention     | pass                  |
| $\omega_H^2(j^2)$ reduced relation                  | Einstein-limit | pass                  |
| each-spin reading of $J$                            | convention     | pass                  |
| critical temperature                                | Einstein-limit | pass                  |
| Wald entropy matches paper                          | Gauss–Bonnet   | pass                  |
| $b'(r_H) = 2r_H/(r_H^2 + \alpha)$ at $\Omega_H = 0$ | Gauss–Bonnet   | pass, $\sim 10^{-14}$ |
| $b', f'$ decrease monotonically in $\Omega_H$       | Gauss–Bonnet   | pass                  |
| <b>published ergosurface radii, their §4.1.1</b>    | Gauss–Bonnet   | <b>fail</b>           |

To these the manuscript adds their near-horizon eqs. (4.11) and (4.12), satisfied to  $9.17 \times 10^{-14}$  and  $6.66 \times 10^{-16}$ , which is what pins the coupling conversion  $\alpha_{\text{ref.}} = 4\alpha$ .

The structure matters more than the tally. The file’s own scope note says the contrasts were chosen among statements “that are closed forms or carry more digits than the section 4.1.1 radii”. Hence:

- the Einstein-limit agreements **do not test the Gauss–Bonnet terms at all**, and the file labels them apart for that reason;
- the two tightest Gauss–Bonnet agreements are against *closed-form* statements — the static horizon-derivative formula at  $\Omega_H = 0$ , which carries no rotation, and the near-horizon algebra, which is computed by `near_horizon.py` and **does not use the bulk rotating solver**;
- the only check that compares our rotating bulk solutions at finite  $\alpha$  against values they obtained *numerically* is the ergosurface radius, which gives 1.102101 against their 1.104 at  $\alpha = 1/2$  in our convention. The manuscript states that the discrepancy is unresolved and declines to offer it as evidence of an erratum.

So, stated strictly: the rotating solver reproduces Myers–Perry exactly at  $\alpha = 0$ , reproduces closed-form EGB statements, and reproduces the near-horizon sector — but its finite- $\alpha$  rotating *bulk* output has one published cross-check, and that one disagrees in the fourth digit.

The manuscript’s containment argument is that the ergosurface radius tests the reconstruction of the metric functions *away from the horizon*, whereas the result depends on the asymptotic charge extraction, the response solve and the zero-temperature extrapolation, none of which uses  $r_e$ . That is a reasonable argument and it is made in the discussion rather than buried; whether it is strong enough is a judgement a referee may well press on.

## Provenance

The per-coupling counts, the  $\alpha$  grid and the temperature ranges were read directly from `results/egb-extremality.json` and reproduce the generated macros `\Nstates = 356`, `\Ncouplings = 13`, `\Nmapcouplings = 6`, `\Nnarrowcouplings = 7` and `\Nmaplist` of `manuscript/numbers.tex`. The description of the walk is read from `src/rotating_bh/extremality.py`.

The description of the rotating solver is read from the docstring of `src/rotating_bh/egb_rotating_bvp.py`; the costs, the five discarded routes and the three errors from `prompts/uso-de-tiempo-y-tokens.md`, sections 2 and 3. The method of [arXiv:1010.0860](#) is from its section 4.1.1 and its reference [41], in the copy under `papers/`. The contrast table is read from `results/egb-rotating-external-contrasts.json`; the ergosurface numbers are from the discussion section of `manuscript/main.tex`.

*Not checked:* no walk was re-run and no solve was performed; the 356 stored rows were not individually re-tested against the acceptance thresholds, and the external contrasts were read from the stored artifact rather than recomputed. The ratio  $508/101.6 = 5.0$  of section 3 is arithmetic on the ledger table; the notes do not show the division.